

# Borel Torsor Descent Obstruction

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## Summary

Pass 91 decides the descent status of the restriction/span Borel torsor over the finite singleton-prime cover site. The global-Levi Borel package is not a sheaf on multi-prime supports. Its sheafification or stackification is the local Loeb object, and the Rosser class is exactly the kernel lost in that process.

For a finite support  $S$ ,

$$P(S) = \left( \prod_{p \in S} \mathbb{Z}_p \right) / \Delta \mathbb{Z}.$$

The singleton-cover descent kernel is

$$\mathbb{Z}^S / \Delta \mathbb{Z} \cong \mathbb{Z}^{|S|-1}.$$

## Main Claims

- **Borel non-separatedness.** The global-Levi Borel prestack  $\mathbb{Q}^\times \ltimes P(S)$  is not a sheaf for  $|S| \geq 2$ .
- **Local Borel stackification.** The local object is

$$(\mathbb{Q}^\times)^S \ltimes \prod_{p \in S} (\mathbb{Z}_p / \mathbb{Z}).$$

- **Rosser kernel.** The horizontal Rosser defect  $\mathbb{Z}^S / \Delta \mathbb{Z}$  is the kernel lost under stackification.
- **Shear action.** The hyperbolic shear action transports global lifts with the same local data but does not choose a canonical zero section.

## Machine Verification

code/scripts/check-pass91.py produced artifacts/reports/pass91-borel-torsor-descent-obstruction-check.json with overall PASS. It checks descent ranks  $|S| - 1$ , finite kernel sizes  $N^{|S|-1}$ , global Borel non-sheaf behavior on multi-prime supports, local-Levi sheafification in a finite proxy, and the fact that shear transports but does not kill descent-kernel lifts.

## Next Use

Pass 92 should relocate the Borel descent obstruction to the Zariski/generic prime site and compare it with the Pass-63  $j_!$  ghost line.